



University of Pittsburgh

The Inapplicability of Nash Equilibrium in The CJT

Why I'd Rather Vote Sincerely and So Should You

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- ▶ What's behind the disagreement?

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Introduction

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CJT

3-Voter
Example

Pivotal
Voting

Decision Theoretic
Nash Equilibrium
Existing Responses

Interpretation

Response

Focal Point

End

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 - ▶ Economists typically assume a positive/predictive interpretation.
 - ▶ I propose a normative interpretation, which makes both arguments relevant.
- ▶ Reject the use of Nash Equilibrium.
- ▶ Propose to use **Focal Point** to understand voter behaviors in large elections.

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- ▶ Condorcet Jury Theorem: If everyone in the group is **competent**, **independent** and **sincere**, plurality voting (simple majority) will converge to truth as the group becomes larger.

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- ▶ Condorcet Jury Theorem: If everyone in the group is **competent**, **independent** and **sincere**, plurality voting (simple majority) will converge to truth as the group becomes larger.
 - ▶ Often used to justify (1) Universal Suffrage, (2) Wisdom of Crowd, (3) Epistemic Benefit of Democracy.

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 - ▶ Objective: Maximize the expected **hit rate** $E[P(\text{Correct})]$.

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- ▶ (4) The candidate with the most votes gets elected, and voters receive their payoffs.

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 - ▶ $\pi^i = L$ if and only if $P^i(s_L | \sigma^i) \geq P^i(s_R | \sigma^i)$.

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 - ▶ Non-asymptotic result justifies **Universal Suffrage**. The more people, the better.
 - ▶ Asymptotic result justifies the **Wisdom of Crowd** and the **Epistemic Benefit for Democracy**.
 - ▶ Read literally, the asymptotic result has little output. We will never have infinitely many voters. But since the convergence is strong, if n is large enough, we are close enough to 1 (or at least closer than other decision making mechanisms).

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- ▶ Clearly separating objective and subjective probabilities is crucial for my discussion on sincere and pivotal voting.

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- ▶ Competence: $P(\sigma_L|s_L) = 0.9$, $P(\sigma_R|s_R) = 0.6$ for everyone.
- ▶ Let's take Bob's perspective. He initially finds s_L and s_R equally likely.

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- ▶ They all start with a common (true) prior $P(s_L) = P(s_R) = 0.5$
- ▶ Competence: $P(\sigma_L|s_L) = 0.9$, $P(\sigma_R|s_R) = 0.6$ for everyone.
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- ▶ Then he receives a signal σ^{Bob} . Suppose that Bob observes $\sigma^{Bob} = \sigma_L$.
- ▶ If Bob is the only voter, he maximizes expected hit rate by voting sincerely, which is L in this case. Overall, sincere voting would deliver the correct answer with probability 0.9 in s_L and with probability 0.6 in s_R . The overall expected hit rate is 0.75.

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Pivotal Voting

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 - ▶ "I do not expect a tie to happen. But if a tie happens in this case, I want to break the tie in the right way by voting for R instead of L ."

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Austen-Smith & Banks (1996)

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- ▶ A fully rational voter should vote **as if she were pivotal**.

Decision Theoretic Argument

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Decision Theoretic Argument

- This table summarizes the decision for Bob.

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- This table summarizes the decision for Bob.

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- ▶ Easy to see this from our example. Bob (or anyone) can benefit from unilateral deviation.
- ▶ Feddersen & Pesendorfer (1997) showed that there are Nash equilibriums where everyone votes pivotally.

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- ▶ My goal: Refuse both arguments and defend **Sincerity**.

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- ▶ But you may ask, if only empirical adequacy matters, is the decision-theoretic argument relevant in any way?

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- ▶ **Normative Interpretation (Advocated Here):** The CJT identifies conditions under which democratic processes (plurality rule) *would* lead to good decisions.
 - ▶ If we want the conclusions to hold, we have *prima facie* reasons to make the assumptions true.

Normative Interpretation

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- ▶ (Conclusion) Everyone should vote sincerely.
- ▶ Under the Normative Interpretation, CJT provides *prima facie* reason for us to make the assumptions true.

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- ▶ The claim that sincere voting is not rational, if true, would undermine the *prima facie* reason to vote sincerely.
- ▶ But it can also be undermined by empirical facts. For example, a group of omniscient agents would make good decisions together. This gives us a *prima facie* reason to be omniscient. But it is overruled by its empirical impossibility. There doesn't seem to be any reason not to be omniscient. It's just empirically impossible.

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- ▶ (2) Normative interpretation shifts our attention from "Does democracy make good decisions?" to "Under what conditions does democracy make good decisions?" I think the latter is a more important question.

Yao Fan

The Heavy Epistemic Burden of NE

Nash Equilibrium Argument

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Nash Equilibrium Argument

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The Heavy Epistemic Burden of NE

Nash Equilibrium Argument

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- ▶ (C1) Real-world voters are unlikely to vote sincerely.
- ▶ (C2) Real-world voters are more likely to vote pivotally than sincerely.
- ▶ Yao: if Nash Equilibrium is not applicable to the CJT, it would deny (P3).

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- ▶ You need to have a subjective probability $P(\pi_L^{-i}|s)$ about others' voting behaviors across different states.
- ▶ Nash Equilibrium implicitly assumes **Common Knowledge of Strategies**. This common knowledge assumption does all the work required (and more) for pivotal voting.
- ▶ By restricting the comparison of sincere voting and pivotal voting to the domain of NE, economists smuggle in this very strong epistemic assumption.

Why Nash Equilibrium is Inapplicable here

Evaluating the Justifications for NE

How is NE usually justified in game theory?

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How is NE usually justified in game theory?

► 1. Evolutionary / Learning Mechanisms

- Requires repeated plays. Elections are paradigmatic one-off interactions.
- Requires *ex post* observation of individual behavior. Secret ballots make this impossible.
- Requires differential payoffs to drive evolution. CJT is a common-interest game; everyone wins or loses together.

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► 2. Rational Strategic Thinking

- Intended for one-off games, but requires all players to possess common knowledge.
- Utterly unrealistic for a voter in a large election with limited cognitive resources.

Critique of the Pivotal Model

Feddersen & Pesendorfer (1997)

Even if we accept NE, the pivotal equilibria (which do aggregate information asymptotically) have severe issues:

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- ▶ **No Pure Strategy Equilibrium:** Voters must rely on complex randomization (mixed strategies) to aggregate information.
- ▶ **Multiplicity:** There are infinitely many symmetric mixed-strategy equilibria. How do millions of voters successfully coordinate on the exact same one without communication?
- ▶ **Non-Informative Voting:** As n grows, Piv^i becomes overwhelmingly informative. The fraction of voters who completely ignore their private signals approaches 1.

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- ▶ **Ambiguity:** Beliefs may exist as intervals (imprecise probability). Calculating Piv^i from endpoints can yield wildly conflicting directives.
- ▶ **Incorrectness:** If a voter holds a precise but incorrect belief, pivotal voting can cause large electorates to elect the wrong candidate with near certainty (cf. Mandler 2012).

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- ▶ I propose to use the concept of the **Focal Point**.
- ▶ Introduced by Schelling (1960) precisely for one-off interactions with common interest under uncertainty.

Focal Point

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Classic Examples

► Coin-Matching Game:

Focal Point

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Classic Examples

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- ▶ **Meeting Game:** Two players must meet in NYC on a given day without prior arrangements. Win if they meet. (Majority chose Grand Central Station at noon).
- ▶ **The Core Idea:** Focal points are options that “stick out” or “suggest themselves” when players are ignorant of others’ strategies.

Coordination in the CJT

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- ▶ However, there is a profound coordination aspect: Will we play as sincere voters or pivotal voters?
- ▶ If voters attempt to vote pivotally but fail to align on the exact same complex mixed-strategy distribution, information aggregation fails. In comparison, sincere voting is more robust.

Focal Point $\neq$ Nash Equilibrium

Bosch-Domènech and Vriend (2013)

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- ▶ **My Claim:** Sincere voting in the CJT is exactly like (Up, Left).

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- ▶ Note: Psychological Salience is only one theory of focal points. There are many others. In the written paper, I went through several different theories and see how sincere voting fares.
- ▶ **Uniqueness:** There are infinitely many pivotal strategies. There is only *one* way to vote sincerely.
- ▶ **Psychological Salience:** It is the “default rule” that directly channels the epistemic spirit of the CJT: contributing your private information to the collective.

Thank You!

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